

ADVERTISEMENT

Exponential Distribution

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Exponential Distribution

Exponential distribution is used for describing time till next event e.g. failure/success etc.

It has two parameters:

scale - inverse of rate (see lam in poisson distribution) defaults to 1.0.

size - The shape of the returned array.

Example

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Draw out a sample for exponential distribution with 2.0 scale with 2x3 size:

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NumPy Random

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```
print(x)
```

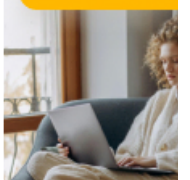
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Visualization of Exponential Distribution

Example

\$1,995 **\$6**

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Uniform Distribution
Logistic Distribution
Multinomial Distribution
Exponential Distribution
Chi Square Distribution
Rayleigh Distribution
Pareto Distribution
Zipf Distribution

NumPy ufunc

ufunc Intro
ufunc Create Function
ufunc Simple Arithmetic
ufunc Rounding Decimals
ufunc Logs
ufunc Summations
ufunc Products
ufunc Differences
ufunc Finding LCM
ufunc Finding GCD

```
from numpy import random
import matplotlib.pyplot as plt
import seaborn as sns

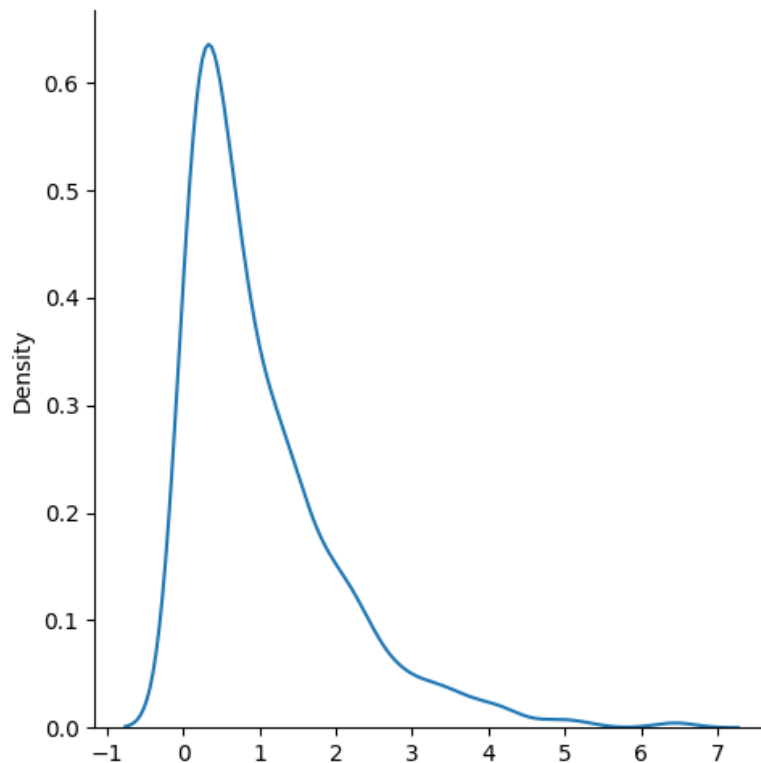
sns.displot(random.exponential(size=1000), kind="kde")

plt.show()
```



R

Result



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Relation Between Poisson and Exponential Distribution

Poisson distribution deals with number of occurrences of an event in a time period whereas exponential distribution deals with the time between these events.

Exercise ?

The `random.exponential()` method has two